

Overview of Linux, Directories and Their Uses

1. Overview of Linux

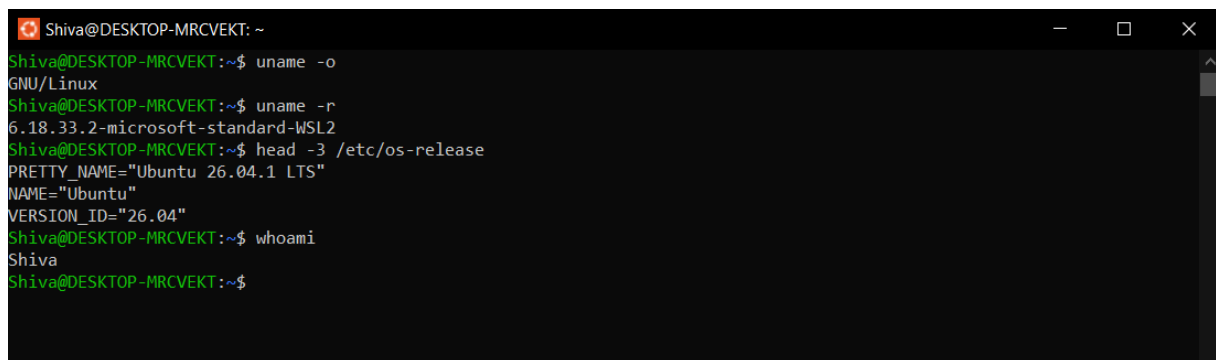
Linux is a free and open-source operating system. An operating system is the software that sits between the hardware of a computer and the programs a person runs on it, and it takes care of memory, storage, processes, devices and user accounts. Linux was started by Linus Torvalds in 1991 and has been developed since then by thousands of contributors around the world.

Strictly speaking, Linux is only the kernel, which is the core part that talks to the hardware. A usable system is the kernel packaged together with system tools, libraries and applications, and such a package is called a distribution. Ubuntu, Debian, Red Hat Enterprise Linux, CentOS, Fedora and Kali Linux are examples of distributions. They all share the same kernel and the same basic layout, so what is learnt on one distribution applies to the others.

Main characteristics

- Open source: the source code is available to read, modify and redistribute free of cost.
- Multi-user and multitasking: many users can log in and many programs can run at the same time.
- Secure and stable: file permissions and user privileges control who can read or change what, and servers can run for long periods without restarting.
- Portable and lightweight: it runs on servers, desktops, mobile phones, routers and small devices.
- Command line driven: most administration work is done through the shell using commands.

The commands below show the basic details of the system used for this assignment, which is Ubuntu 26.04.1 LTS running under WSL2 on a Windows machine.

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~'. The terminal shows the following commands and their outputs:

```
Shiva@DESKTOP-MRCVEKT:~$ uname -o
GNU/Linux
Shiva@DESKTOP-MRCVEKT:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
Shiva@DESKTOP-MRCVEKT:~$ head -3 /etc/os-release
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
Shiva@DESKTOP-MRCVEKT:~$ whoami
Shiva
Shiva@DESKTOP-MRCVEKT:~$
```

Screenshot 1: Checking the operating system, the kernel version and the logged-in user.

2. Going Through the Directories

Windows keeps files under separate drive letters such as C: and D:. Linux does not work that way. It has a single tree that starts at one point called the root directory, written as a single forward slash (/). Every disk, partition, USB drive and network share is attached somewhere inside that one tree, so there is only ever one starting point to remember.

The layout of this tree is not decided by each distribution on its own. It follows an agreed standard called the Filesystem Hierarchy Standard, which is the reason configuration files are found in /etc on almost every Linux system. The command `ls /` lists the directories present at the top of the tree.

```
Shiva@DESKTOP-MRCVEKT: /
Shiva@DESKTOP-MRCVEKT:~$ cd /
Shiva@DESKTOP-MRCVEKT:/$ pwd
/
Shiva@DESKTOP-MRCVEKT:/$ ls
bin  dev  home  lib  lost+found  mnt  proc  run  snap  sys  usr
boot  etc  init  lib64  media  opt  root  sbin  srv  tmp  var
Shiva@DESKTOP-MRCVEKT:/$
```

Screenshot 2: Moving to the root directory and listing the directories directly under it.

Each of these top-level directories has a defined job. The table below lists the ones seen most often. A listing may also show one or two extras, such as /snap or /init, that belong to the particular setup being used.

Directory	What it is used for
/	The root of the whole filesystem. Every other directory branches from here.
/bin	Essential user commands such as ls, cp, mv and cat.
/sbin	System administration commands, normally used by the root user.
/boot	Files needed to start the system, including the Linux kernel and bootloader.
/dev	Device files that represent hardware such as disks, USB drives and terminals.
/etc	System-wide configuration files for the operating system and installed software.
/home	Personal directories for normal users, for example /home/shivakumar.
/root	The home directory of the root (administrator) user.
/lib	Shared libraries required by the programs in /bin and /sbin.
/media	Mount point where removable media such as USB drives and CDs appear.
/mnt	Mount point used to attach filesystems temporarily by hand.
/opt	Optional add-on software installed separately from the distribution.
/proc	A virtual directory that shows live information about running processes.
/sys	A virtual directory that exposes information about devices and the kernel.
/run	Runtime data created since the last boot, such as process ID files.
/srv	Data served by the system, for example files for a web or FTP server.
/tmp	Temporary files, usually cleared automatically when the system restarts.
/usr	Programs, libraries and documentation installed for all users.
/var	Files that keep changing in size, such as log files, mail and print queues.

Looking inside a few of them shows this clearly. /etc holds configuration files, /bin holds the commands that are typed every day, and /var/log holds the log files that record what the system has been doing.

```
Shiva@DESKTOP-MRCVEKT: /
Shiva@DESKTOP-MRCVEKT:/$ ls /etc | head -5
PackageKit
X11
adduser.conf
alternatives
apparmor
Shiva@DESKTOP-MRCVEKT:/$ ls /bin | head -5
NF
X11
[
aa-enabled
aa-exec
Shiva@DESKTOP-MRCVEKT:/$ ls /var/log
README      auth.log      chrony        dpkg.log      kern.log      private      wtmp
alternatives.log  bootstrap.log dist-upgrade  fontconfig.log landscape     syslog
apt          bttmp        dmesg         journal       lastlog      unattended-upgrades
Shiva@DESKTOP-MRCVEKT:/$
```

3. Why Are There Directories and What They Are Used For

A directory, also called a folder, is simply a container that holds files and other directories. A working Linux system has hundreds of thousands of files, and without directories they would all sit in one place with no way to tell them apart. Directories exist for the following reasons.

Organisation and easy searching

Related files are kept together, so a file can be found by following a path instead of searching the whole disk. Configuration goes to /etc, logs go to /var/log, and personal work goes under /home. The system and the user both know where to look.

Separation of system files and user files

Files belonging to the operating system are kept apart from files created by users. A mistake made while working in a personal folder cannot damage the programs the system needs to boot, and reinstalling the system does not have to touch personal data.

Security and permissions

Permissions in Linux are applied to directories as well as to files. Because system directories such as /etc and /sbin are owned by root, an ordinary user can read what is needed but cannot change or delete it. Each user's own directory under /home is private to that user.

Support for multiple users

Every user gets a separate directory under /home, so several people can use the same machine without seeing or overwriting each other's work.

Mounting devices and backups

Directories such as /media and /mnt act as the points where external disks and network drives are attached to the tree. Grouping data this way also makes backups simple, since copying /home saves all user data in one step.

A predictable structure

Because the same structure is used across distributions, an administrator moving from Ubuntu to Red Hat already knows where the configuration, logs and commands are kept. Scripts written on one system therefore work on another.

The screenshot below shows a personal home directory, including a folder created to keep the notes and scripts of this Linux class separate from everything else.



```
Shiva@DESKTOP-MRCVEKT: ~  
Shiva@DESKTOP-MRCVEKT:~$ cd ~  
Shiva@DESKTOP-MRCVEKT:~$ pwd  
/home/Shiva  
Shiva@DESKTOP-MRCVEKT:~$ ls  
linux_class  
Shiva@DESKTOP-MRCVEKT:~$ ls linux_class  
notes  scripts  
Shiva@DESKTOP-MRCVEKT:~$ ls linux_class/notes  
day1_overview.txt  day2_directories.txt  
Shiva@DESKTOP-MRCVEKT:~$
```

Screenshot 4: The home directory and a folder created to organise the notes and scripts of this class.